

# Yu Wang

Assistant Professor | Southwest Jiaotong University

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<https://mrgodfrey.github.io>

## Research Profile

My research concerns fundamental questions in control theory and inverse problems for partial differential equations. I am particularly interested in how controllability, observability, unique continuation, and identifiability are preserved, weakened, or lost under randomness, discretization, and partial observation.

A major part of my work studies the transfer of control and observability mechanisms across different mathematical settings. This includes understanding when deterministic control structures can be lifted to stochastic systems, and when continuous estimates and unique continuation properties remain valid, uniformly and quantitatively, after numerical discretization.

On the inverse-problem side, I study what can be recovered from limited or indirect observations, with emphasis on identifiability and quantitative stability for unknown sources and related structures. More broadly, I am interested in the mathematical boundary between what information a PDE system retains and what is irretrievably lost.

Research interests: Control Theory; Inverse Problems; Stochastic PDEs

## Appointments

### Assistant Professor

06.2023 - present

School of Mathematics, Southwest Jiaotong University

### Visiting Scholar

09.2025 - 02.2026

Chair for Dynamics, Control, Machine Learning and Numerics, FAU Erlangen-Nürnberg

Host Professor: Prof. Dr. DhC. Enrique Zuazua

## Education

### Ph.D. in Mathematics

09.2017 - 06.2023

Sichuan University

Supervisor: Professor Qi Lü

### B.Sc. in Mathematics

09.2013 - 06.2017

Sichuan University

## Publications

### Books

#### 1. Inverse Problems for Stochastic Partial Differential Equations

Q. Lü, Y. Wang

SpringerBriefs on PDEs and Data Science, Springer Nature Singapore, 2026. ISBN 9789819590476 [Book](#) | [ArXiv](#) | [Blog](#)

### Journal Articles

#### 1. Exact controllability for stochastic first-order multi-dimensional hyperbolic systems

Z. Li, Q. Lü, Y. Wang, H. Yang

SIAM J. Control Optim., Accepted [ArXiv](#) | [Blog](#)

## **2. Null controllability for stochastic fourth order semi-discrete parabolic equations**

**Y. Wang**, Q. Zhao

Appl. Math. Optim., Accepted [ArXiv](#) | [Blog](#)

## **3. scDCL: A multi-view single-cell RNA sequencing clustering method based on dual contrastive learning**

L. Gan, H. Meng, Y. Chen, **Y. Wang**

Computational Biology and Chemistry, (2026), 108998 [Article](#) | [Blog](#)

## **4. An Inverse Source Problem for Semilinear Stochastic Hyperbolic Equations**

Q. Lü, **Y. Wang**

Inverse Problems, 41(11) (2025), 115014 [Article](#) | [ArXiv](#) | [Blog](#)

## **5. The $\phi$ -null controllability for semi-discrete stochastic semilinear parabolic equations**

**Y. Wang**, Q. Zhao

ESAIM Control Optim. Calc. Var., 31 (2025), Paper No. 98 [Article](#) | [ArXiv](#) | [Blog](#)

## **6. Exact controllability for a refined stochastic plate equation**

Q. Lü, **Y. Wang**

Chinese Ann. Math. Ser. B, 46(3) (2025), 415-442 [Article](#) | [ArXiv](#) | [Blog](#)

## **7. Stability and regularization for ill-posed Cauchy problem of a stochastic parabolic differential equation**

F. Dou, P. Lü, **Y. Wang**

Inverse Problems, 40(11) (2024), 115005 [Article](#) | [ArXiv](#) | [Blog](#)

## **8. Null controllability for stochastic coupled systems of fourth order parabolic equations**

**Y. Wang**

J. Math. Anal. Appl., 538 (2024), 128426 [Article](#) | [ArXiv](#) | [Blog](#)

## **9. Null controllability for fourth order stochastic parabolic equations**

Q. Lü, **Y. Wang**

SIAM J. Control Optim., 60 (2022), 1563-1590 [Article](#) | [ArXiv](#) | [Blog](#)

## **Preprints**

### **1. Null Controllability of a Stochastic Parabolic Equation with a Space-Dependent Analytic Noise Coefficient**

Q. Lü, **Y. Wang**

Submitted [Blog](#)

### **2. Stability Estimates for the Inverse Recovery of Drift and Diffusion Point Sources in Stochastic Parabolic Equations**

Q. Lü, **Y. Wang**

Submitted [ArXiv](#) | [Blog](#)

### **3. Carleman Estimates for Wave Equations on the Half-Line: AI-Assisted Weights and Applications**

C. An, **Y. Wang**, Q. Ye, Z. Zhang, Q. Zhuang

Submitted [ArXiv](#) | [Blog](#)

### **4. A Geometric Inverse Source Problem for Stochastic Parabolic Equations**

Y. Li, Q. Lü, M. Qian, **Y. Wang**

Submitted [ArXiv](#) | [Blog](#)

Research Grants

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Controllability of stochastic plate equations</b><br>Sichuan Science and Technology Program (No. 2026NSFSC0777)<br>Funding: RMB 100K                                         | 2026-2027 |
| <b>Controllability of fourth order semilinear stochastic partial differential equations</b><br>National Natural Science Foundation of China (No. 12401589)<br>Funding: RMB 300K | 2025-2027 |
| <b>Controllability of a fourth-order coupled stochastic parabolic system</b><br>Fundamental Research Funds for the Central Universities (No. 2682024CX013)<br>Funding: RMB 100K | 2024-2025 |

Teaching

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|------------------------------------------|-----------|
| <b>Neural Networks and Deep Learning</b> | 2025-2026 |
| <b>Introduction to Major</b>             | 2024      |

Academic Service

Mathematical Reviews / MathSciNet (American Mathematical Society) - reviewer

Journal refereeing

Applied Mathematics & Optimization; Asian Journal of Control; Chinese Annals of Mathematics, Series B; ESAIM: Control, Optimisation and Calculus of Variations; IMA Journal of Mathematical Control and Information; Inverse Problems; Journal of Differential Equations; Journal of Mathematical Analysis and Applications; Journal of Optimization Theory and Applications; Journal of Systems Science and Complexity; Mathematical Control and Related Fields; Neural Networks; Science China Mathematics; SIAM Journal on Control and Optimization; Systems & Control Letters.